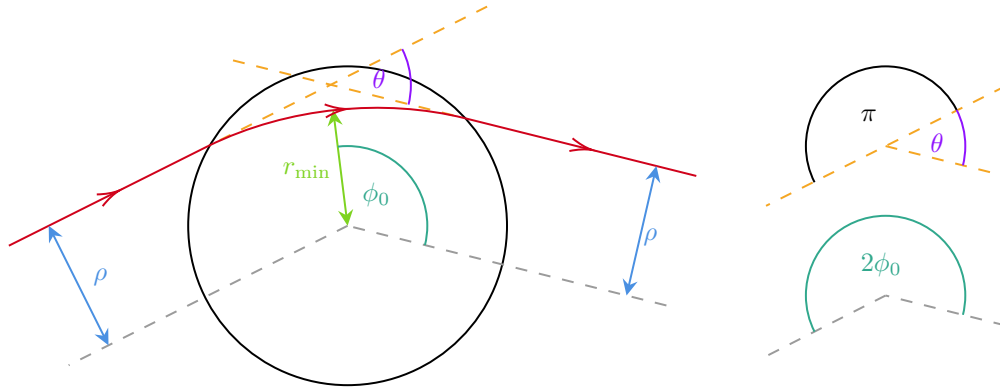


Ejercicio 2b

Simon Fonseca

May 6, 2026



From part (a), we found the potential:

$$U(r) = \begin{cases} 2E_0 \left(1 - \frac{R}{r}\right), & r < R \\ 0, & r \geq R \end{cases}$$

Now we're shooting particles at a different energy E and need to find the scattering cross-section. The scattering angle θ relates to the angle ϕ_0 through:

$$\theta = \pi - 2\phi_0$$

where ϕ_0 is the angle from the initial direction to the point of closest approach.

The angle ϕ_0 comes from integrating the trajectory:

$$\phi_0 = \int_{\infty}^{r_{min}} \frac{M dr}{mr^2 \sqrt{\frac{2}{m}(E - U(r)) - \frac{M^2}{m^2 r^2}}}$$

where $M = mv_{\infty}\rho$ and $E = \frac{1}{2}mv_{\infty}^2$.

This splits into two regions because $U(r)$ has two forms:

$$\phi_0 = \int_{r_{min}}^R \frac{\rho/r^2 dr}{\sqrt{1 - \frac{4E_0}{mv_{\infty}^2} \left(1 - \frac{R}{r}\right) - \frac{\rho^2}{r^2}}} + \int_R^{\infty} \frac{\rho/r^2 dr}{\sqrt{1 - \frac{\rho^2}{r^2}}}$$

The second integral (outside the potential) is simple:

$$\int_R^\infty \frac{\rho/r^2 dr}{\sqrt{1 - \frac{\rho^2}{r^2}}} = \arctan\left(\frac{\rho}{\sqrt{R^2 - \rho^2}}\right)$$

Inside the potential region, notice:

$$U(r) = 2E_0 - \frac{2E_0 R}{r}$$

The constant $2E_0$ just shifts the energy. The $-\frac{2E_0 R}{r}$ term is exactly like a Coulomb potential with $\alpha = 2E_0 R$.

So inside $r < R$, the trajectory is a conic section (like Kepler orbits):

$$\frac{p}{r} = 1 + e \cos \phi$$

with parameters:

$$p = \frac{M^2}{2mE_0 R}, \quad e = \sqrt{1 + \frac{2(E - 2E_0)M^2}{4mE_0^2 R^2}}$$

meaning that, at the closest approach ($r = r_{min}$), the radial velocity is zero, which occurs when $\phi = 0$ (by our choice of coordinates). So:

$$\frac{p}{r_{min}} = 1 + e \Rightarrow r_{min} = \frac{p}{1 + e}$$

The integral inside the potential region goes from r_{min} to R :

$$\phi_0^{(inside)} = \int_{r_{min}}^R \frac{\rho/r^2 dr}{\sqrt{1 - \frac{4E_0}{mv_\infty^2} \left(1 - \frac{R}{r}\right) - \frac{\rho^2}{r^2}}}$$

Since we know the trajectory is a conic section with $\frac{p}{r} = 1 + e \cos \phi$, we can change variables from r to ϕ :

From $r = \frac{p}{1 + e \cos \phi}$, we get:

$$dr = \frac{pe \sin \phi}{(1 + e \cos \phi)^2} d\phi$$

The integral becomes:

$$\phi_0^{(inside)} = \int_0^{\phi_R} d\phi$$

where ϕ_R is the angle when the particle exits at $r = R$:

$$\frac{p}{R} = 1 + e \cos \phi_R$$

So:

$$\cos \phi_R = \frac{1}{e} \left(\frac{p}{R} - 1 \right)$$

Outside the potential:

$$\phi_0^{(outside)} = \arctan \left(\frac{\rho}{\sqrt{R^2 - \rho^2}} \right)$$

This is the additional angle traveled from $r = R$ to $r = \infty$ along a straight line.
Total ϕ_0

$$\phi_0 = \phi_R + \arctan \left(\frac{\rho}{\sqrt{R^2 - \rho^2}} \right)$$

where:

$$\phi_R = \arccos \left[\frac{1}{e} \left(\frac{p}{R} - 1 \right) \right]$$

Now substitute back, you get:

$$\tan \phi_0 = \frac{\rho(mv_\infty^2 - 2E_0)}{2E_0\sqrt{R^2 - \rho^2}}$$

Since $\phi_0 = \frac{\pi - \theta}{2}$, we have $\tan \phi_0 = \cot(\theta/2)$:

$$\rho = \frac{2E_0 R \cot(\theta/2)}{\sqrt{4E_0^2 + m^2 v^4 - 4E_0 m v^2 + 4E_0^2 \cot^2(\theta/2)}}$$

The differential cross-section is:

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\pi \sin \theta} \cdot 2\pi \rho \left| \frac{d\rho}{d\theta} \right| = \frac{1}{\sin \theta} \rho \left| \frac{d\rho}{d\theta} \right|$$

After computing $\frac{d\rho}{d\theta}$, you get:

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{E_0^2 R^2}{(E_0 - E)^2} [E_0^2 \cos^2(\theta/2) + (E_0 - E)^2 \sin^2(\theta/2)]^{-2}}$$

where $E = \frac{1}{2}mv^2$.

As $E \rightarrow E_0$:

$$\frac{d\sigma}{d\Omega} \rightarrow \frac{E_0^2 R^2}{(E_0 - E)^2 \cdot E_0^4 \cos^4(\theta/2)} \sim \frac{R^2}{(E_0 - E)^2 \cos^4(\theta/2)}$$

The cross-section diverges. This makes physical sense: at $E = E_0$, all particles with $\rho < R$ scatter to exactly $\theta = \pi$, creating a singularity.